

1. Write prototype of a function that takes String as argument and returns int
2. Design a class to overload a function Series as follows:

void Series(int x, int n)

$$S = \frac{x}{2} - \frac{2x}{3} + \frac{3x}{4} - \frac{4x}{5} + \frac{5x}{6} - \dots + \frac{nx}{n+1}$$

void Series (int n)

$$S = \frac{2^{3!}}{4!} + \frac{3^{4!}}{5!} + \frac{4^{5!}}{6!} + \frac{5^{6!}}{7!} + \dots + \frac{n^{(n+1)!}}{(n+2)!}$$

void Series()

$$S = 2! + 3! + 5! + 7! + 11! + 13! + 17! + \dots + n!$$

3. Write a program to overload a function pattern as follows:
void pattern()

```
* * * * *
*           *
*           *
*           *
*           *
* * * * *
```

void pattern(int n)

```
  *
 * *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

→ if n == 4

void pattern(char ch)

```
* * * * *
 * * * *
  * * *
   * *
    *
   * *
  * * *
 * * * *
* * * * *
```

→ if ch == '*' (printing ch)

4. Design a class to overload a function Numbers as follows:

void Numbers()

```

1 * * * *
* 2 * * *
* * 3 * *
* * * 4 *
* * * * 5
  
```

void Numbers(int n)

```

1 2 3 4 5
1 2 3 4 5
1 2 3 5 4
1 2 5 4 3
1 5 4 3 2
  
```

void Numbers(char ch)

```

1 * * * *
2 2 * * *
3 3 3 * *
4 4 4 4 *
5 5 5 5 5
  
```

if ch == '*'

void Numbers(char ch1, char ch2)

```

1 2 1 2 1
2 1 2 1 2
1 2 1 2 1
2 1 2 1 2
  
```

5. Design a class to overload a function loop as follows

loop()

```

*   O   O   O   O   #
O   #   O   O   *   O
O   O   *   #   O   O
O   O   *   #   O   O
O   #   O   O   *   O
*   O   O   O   O   #
  
```

loop(int n) [here n is the number of lines] to print Pascal's Triangle

```

    1
  1 1
 1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1
```

** Print n where n starts from 1 in each row and keeps updating like $n = n * (i - j) / j$

6. Define a class to overload a function Sum() as follows:

Sum()

$$S = \frac{\sqrt[3]{2}^{\sqrt{4}}}{1} - \frac{\sqrt[4]{3}^{\sqrt{5}}}{1+2} + \frac{\sqrt[5]{4}^{\sqrt{6}}}{1+2+3} - \frac{\sqrt[6]{5}^{\sqrt{7}}}{1+2+3+4} + \dots \text{ upto } n\text{th term}$$

Sum(int n)

Find $\sqrt{\sqrt{\sqrt{\dots \sqrt{n}}}}$ [square root of n for n times]

Sum(long k) print Fibonacci as right-angled triangle pattern till k

```

1
2 3
5 8 13
21 34 55 89 ...k
```

1. Define a class as follows:

Class name: Bill

Data members:

String cname; → to store consumer name

int units; → Units consumed

double amount → to store calculated amount

Member methods:

void accept() → to accept name and units consumed from user

void calculate() → to calculate amount according to the tariff

void display() → to print display name and amount

Write the driver code/main function to call the user defined functions

Units	rate
0 to 100	5/unit
101 to 300	7/unit
Above 300	10/unit

2. Define a class as follows:

Class name: Interest

Data members: int p, t, r (for principal, rate and time)
double amt (for amount)

Member method:

void input() to take p, t and r from user

void compute() to find total amount and store in amt

void show() to show the amount

Write the main function to call above functions.

$$\text{amt} = p (1 + r/100)^t$$

3. Define a class as follows:

Class name: WaterBill

Data members:

String cname; → to store consumer name

Int waterUsage; → Units consumed

double amount → to store calculated amount

Member methods:

void accept() → to accept name and water consumed from user

void calculate() → to calculate amount according to the tariff

void display() → to print display name and amount

Write the driver code/main function to call the user defined functions

Units	Rate per 100 liters
0 to 50	Rs.2
51 to 200	Rs.3
Above 200	Rs.5

